Reactions to the Gillette "We Believe" Advertisement on r/MensLib and r/MensRights

Methodology, Statistics and Replication Materials

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This document records how the corpus was collected, how tone and stance were measured, how the statistics were computed, and what ships in the replication package. It is a companion to the findings write-up.

1 Study design

Posts and comments about the Gillette "We Believe: The Best Men Can Be" advertisement (released 14 January 2019) were collected from two subreddits, r/MensLib and r/MensRights, over January 2019 – September 2026, and measured on two dimensions:

- **Tone** — affective valence of the text, from local transformer and lexicon models.
- **Stance and constructs** — position on the ad, plus resentment, identity threat, zero-sum thinking and hostility, from LLM coding.

The two dimensions answer different questions, and the distinction carries the study. Tone measures how a comment sounds; stance measures what position it takes. A warm-sounding r/MensRights comment praising the Egard rebuttal video is positive in tone and critical in stance. Tone differs little between the communities and its sign is not identified once differential moderation is accounted for; stance differs enormously and survives the same sensitivity analysis.

2 Data collection

Source. The Arctic Shift Reddit archive (`arctic-shift.photon-reddit.com`), a free public mirror requiring no API key. Reddit's own API caps search at roughly 1,000 results and skews toward recent content, which makes it unusable for a January 2019 event; Pushshift, the standard alternative, is now restricted to subreddit moderators.

Complete post recall. Arctic Shift's keyword search times out on a subreddit as large as r/MensRights. Rather than accept partial results, the scraper listed *every* post in both subreddits month by month — 171,723 posts — and filtered locally against the relevance rule. Post recall is therefore a census of the two subreddits over the window, not a search result.

Two comment sweeps, merged and de-duplicated, with provenance recorded in the flags `from_thread` and `from_kw_search`:

1. *Thread sweep* — every comment in a thread whose post matches, whether or not the comment names the ad. This preserves conversation: 13,856 comments never mention the ad but reply within those threads.
2. *Keyword sweep* — comments mentioning Gillette elsewhere in the same two subreddits, catching the ad invoked in later, unrelated discussions.

Search terms. `Gillette` plus the misspellings `Gilette` and `Gillete`. "We Believe" could not be searched as a phrase, since the server matches the two words separately, so it is matched locally in the retrieved text.

Relevance rule. A unit is relevant if it matches the strict phrases (Gillette ad / commercial / advertisement; "We Believe" ad / commercial / advertisement), *or* mentions Gillette alongside an ad-context word (ad, commercial, campaign, boycott, toxic masculinity, P&G), *or* mentions Gillette between 13 January and 30 June 2019, when any such mention concerns the ad. Comments

also qualify by sitting in a relevant thread. The unrelated Moore & Gillette book on masculinity is excluded by name.

Verification. Comments retrieved per thread were reconciled against each post's own `num_comments` field: median 100 % per thread, 105 % in aggregate, the excess because the archive also holds comments posted after that count was recorded. No duplicate IDs. Live spot-checks against reddit.com were not possible — Reddit blocks automated fetching from the collection machine — so every row stores a permalink for manual verification.

3 Sample construction

Every one of the 17,643 collected units carries a status. Nothing is deleted from the dataset; excluded rows remain with a flag, so all counts reconcile.

Table 1: Sample construction

	Collected	Removed	Bots	Mod notices	No text	**Analysed**
r/MensLib comments	1,743	390 (22.4 %)	2	12	4	**1,335**
r/MensRights comments	15,280	755 (4.9 %)	24	3	109	**14,389**
Posts (both subs)	620	0	0	0	1	**619**

Exclusion rules. Removed or deleted bodies (1,145) store the literal string `[removed]`, so no text exists to measure; this is a property of the data rather than a judgement, and the differential rate across subreddits is itself a result. Units with no codeable text (114) are bare URLs, bare markdown links and single characters — note that very short comments are *kept*, since "#boycottgillette" carries stance. Bots (26) are known bot accounts plus bodies containing "I am a bot". Moderator notices (15) are removal announcements, which are moderation actions rather than opinions.

Posts whose body was removed are retained (100 cases), because the title survives and carries codeable text. This is why 619 of 620 posts are analysed.

Normalisation is deliberately minimal: HTML entity decoding, zero-width character removal, whitespace collapsing. No stemming, stopword removal or case folding, since the transformer and LLM measures expect natural text. 122 units share identical text; each distinct text was coded once and the label copied to its duplicates.

4 Measures

4.1 Tone

Table 2: Tone measures

Measure	Model	Output
Polarity (primary)	`cardiffnlp/twitter-roberta-base-sentiment-latest`	$p(\text{pos}) - p(\text{neg})$, $[-1, 1]$
Emotion	`j-hartmann/emotion-english-distilroberta-base`	7 probabilities; argmax
Polarity (robustness)	VADER 3.3.2	compound, $[-1, 1]$

Truncation is at 256 word-piece tokens, not 128. This matters: 128 would truncate about 15 % of comments, and truncation is asymmetric across subreddits because r/MensLib comments are twice as long, which would manufacture the very difference being tested.

4.2 Stance and constructs

`gpt-4.1-mini`, temperature 0, one call per unit, OpenAI structured outputs under a strict JSON schema. 16,221 unique texts, zero failures, $8.90, 25 minutes.

The model returns one `ad_stance` label (supportive, critical, mixed, not_about_ad), four constructs rated 0–3 (0 absent, 1 mild or implicit, 2 clearly present, 3 strong and central), a confidence, and a

rationale of at most fifteen words retained for auditing. The construct definitions as given to the model:

- **resentment** — grievance or bitterness that the author's group is treated unfairly, taken for granted or wronged. Not mere disagreement: it carries a sense of injustice done to "us".
- **identity threat** — sense that being a man, or masculinity itself, is devalued, attacked, pathologised or misrepresented.
- **zero-sum** — frames one group's gains as necessarily another's losses.
- **hostility** — antagonism toward people or groups: insults, contempt, dehumanising language, mockery aimed at persons. Anger alone does not count unless directed at people.

Six rules govern the hard cases: code only the author's own words, since lines beginning ">" are quoted from someone else; judge position, not tone; attacking the ad's *critics* is supportive while attacking its *defenders* is critical; a reproduced news headline with no added opinion is not_about_ad, but a headline carrying the sharer's sarcasm or an added tag is coded; boycotting Gillette or buying a rival brand *because of the ad* is critical; and a stance is coded only when the comment itself refers to the ad, never inferred from the thread's topic.

The last three rules were added after a 200-unit pilot exposed two systematic errors: neutral news headlines read as endorsements, and stance inferred from thread topic for deep reply-chain comments that never mention the ad. Adding them raised blind agreement from $\kappa = 0.52$ to $\kappa = 0.72$. Pilot codings were discarded rather than merged, so the released corpus is internally consistent under the final prompt. The verbatim system prompt and JSON schema are in `analysis/prompt_verbatim.md`.

4.3 Themes

MiniLM-L6-v2 embeddings (mean-pooled, L2-normalised, 256 tokens), reduced by truncated SVD to 50 dimensions, clustered with k-means, k chosen by silhouette over 8–20 ($k = 8$). Clusters were labelled by reading the top c-TF-IDF terms and six nearest-centroid comments per cluster from both subreddits. Silhouette peaks at 0.077, so these are soft groupings and should be reported as rough and interpretive rather than as recovered latent structure.

5 Inference

Comments nest in threads, and the nesting is extreme: five threads carry 83 % of r/MensLib's comments, across 1,205 threads in total. Treating 15,724 comments as independent would produce standard errors far too small, so every interval accounts for clustering.

- **Primary tone specification.** OLS of polarity on an r/MensRights indicator with $\log(1 + \text{words})$, a top-level indicator and period fixed effects, with thread-clustered standard errors, plus a wild cluster bootstrap (Rademacher weights, 2,000 replications). The bootstrap matters because treatment is constant within cluster and the cluster-size distribution is extreme, conditions under which asymptotic cluster-robust standard errors are too narrow.
- **Unadjusted differences.** Bootstrap resampling whole threads, 2,000 replications.
- **Stance.** Logistic regression of *critical* on the same covariates with thread-clustered standard errors, alongside the thread-bootstrapped difference in critical share.
- **Effect sizes.** Raw difference, Cohen's d and Cliff's δ. Cliff's δ is preferred for polarity because the distribution is bimodal at ± 1, which makes a pooled-SD d misleading.

Comment score is never used as a control: it is downstream of the comment and would be a collider. Length is in every specification because longer comments score more negative — polarity falls from -0.04 in the shortest decile to -0.50 in the longest — and r/MensLib comments are twice as long, so length suppresses the raw difference.

Differential moderation. r/MensLib removed 22.4 % of its comments against r/MensRights' 4.9 %, and the text is unrecoverable, which is selective attrition directly on the outcome. Lee bounds equalise retention by trimming 19.3 % from the lower-removal group's tails; Manski

bounds assign the unobserved comments their most adverse values. The resulting asymmetry is the central methodological finding.

Table 3: Sensitivity to differential moderation

	Observed	Adjusted	Lee bounds	Manski bounds	Sign identified?
Tone: polarity gap	$+0.044$	$+0.123$	$[-0.086, +0.249]$	$[-0.193, +0.384]$	No
Stance: critical gap	-0.734	Table 6	—	$[-0.497, -0.770]$	Yes

Specification curves re-estimate both results across date windows, provenance subsets, confidence thresholds, top-level-only comments, exclusion of the 21 authors who post in both subreddits, and downsampling r/MensRights. The tone gap stays positive but small throughout; the stance gap stays between -70 and -75 points.

6 Results

Table 4: Descriptive statistics, analysis sample

	r/MensLib	r/MensRights
Comments analysed	1,335	14,389
Unique authors	549	4,846
Threads	81	1,124
Median words	50	25
Top-level	17.5 %	34.2 %
January–February 2019	68.9 %	61.3 %
Mean polarity (SD)	-0.335 (0.506)	-0.379 (0.516)
Mean VADER	$+0.056$	-0.023
Takes a position on the ad	38.2 %	42.0 %
of those, critical	12.9 %	86.4 %

Table 5: Tone: OLS with thread-clustered standard errors. Outcome is RoBERTa polarity except in column (5). The coefficient is on the r/MensRights indicator, so negative means more negative than r/MensLib.

	(1)	(2)	(3)	(4)	(5) VADER
r/MensRights	-0.044	-0.123	-0.119	-0.123	-0.125
Cluster SE	0.032	0.028	0.028	0.029	0.042
t	-1.39	-4.37	-4.26	-4.20	-2.95
p	0.165	1.3×10^{-5}	2.1×10^{-5}	2.7×10^{-5}	0.003
log(words)	—	-0.121	-0.121	-0.120	-0.065
Top-level	—	—	yes	yes	yes
Period FE	—	—	—	yes	yes
R^2	0.001	0.069	0.069	0.070	0.019

$N = 15{,}724$ comments in 1,205 thread clusters throughout. Column (1) versus (2) is the length-suppression result: the difference is indistinguishable from zero until comment length is controlled. Wild cluster bootstrap $p < 0.001$ for column (4).

Table 8 separates community from position. Resentment and identity threat track criticism of the ad rather than subreddit membership: critical comments average 1.83 and 1.61 wherever they were posted, against 0.43 and 0.44 for supportive ones.

Table 6: Stance on the advertisement

	Supportive	Mixed	Critical	Not about the ad
r/MensLib, count	366	78	66	825
r/MensRights, count	660	164	5,218	8,347
r/MensLib, % of positioned	71.8	15.3	12.9	—
r/MensRights, % of positioned	10.9	2.7	86.4	—

Logit: comment is critical, among 6,552 positioned comments, 1,010 clusters	
r/MensRights log-odds (SE)	3.552 (0.182)
Odds ratio [95 % CI]	**34.9** [24.41, 49.88]
p	1.3×10^{-84}
Raw difference in critical share	−0.734

Table 7: Constructs by subreddit. Ratings 0–3; confidence intervals resample whole threads.

	Resentment	Identity threat	Zero-sum	Hostility
r/MensLib, mean (SD)	0.333 (0.573)	0.340 (0.717)	0.057 (0.253)	0.122 (0.378)
r/MensRights, mean (SD)	0.908 (1.037)	0.747 (1.075)	0.451 (0.730)	0.416 (0.779)
≥ 2, r/MensLib	5.2 %	10.4 %	0.5 %	1.5 %
≥ 2, r/MensRights	29.8 %	23.6 %	11.2 %	10.1 %
Difference	−0.575	−0.407	−0.394	−0.294
95 % CI	[−0.663, −0.475]	[−0.572, −0.238]	[−0.439, −0.341]	[−0.327, −0.265]
Cohen's d	−0.572	−0.388	−0.561	−0.391
Cliff's δ	−0.292	−0.192	−0.276	−0.178

7 Validation

Labels were assigned without sight of model output, on stratified samples spanning both subreddits and the full score range. Both labelled samples ship with the package, each carrying the model's label alongside, so κ is recomputable and a further coder can work through them blind.

Three findings follow. First, the error patterns are interpretable: VADER called 9 of 24 negatives positive, defeated by the sarcasm abundant in this corpus, while RoBERTa's opposite weakness labelled 10 of 27 neutrals negative, reading procedural remarks as hostile. Second, measurement error does not favour either side — RoBERTa accuracy was 75.0 % in r/MensLib against 70.0 % in r/MensRights — which is the property that matters for a between-community comparison. Third, prompt wording moved κ by 0.20, from 0.52 to 0.72, which quantifies how much LLM coding depends on instruction design and argues for publishing the prompt verbatim rather than describing it.

The stance coding was additionally checked on the subsets most likely to break it — quote-heavy comments, very short comments and low-confidence codings — with no material change.

8 Replication materials

The package is `Gillette_Replication_Package.zip`, with `REPLICATION.md` as its entry point and `MANIFEST.csv` listing every file with size and SHA-256 prefix. Each pipeline step is resumable; steps 6–10 are deterministic under fixed seeds.

Contents. Code: `scrape.py`, `summary.py`, seven analysis modules, the build scripts, the verbatim prompt, and `requirements-frozen.txt` (Python 3.13.5; pandas 2.3.3, numpy 2.3.1, scipy 1.16.0, scikit-learn 1.8.0, statsmodels 0.15.0, torch 2.6.0, transformers 5.3.0, openai 2.9.0, vaderSentiment 3.3.2). Data: the full comment and post tables with all statuses, the cleaning statuses, tone scores

Table 8: Constructs by coded stance, pooling both subreddits

Stance	n	Resentment	Identity threat	Zero-sum	Hostility
Critical	5,284	1.830	1.610	0.982	0.698
Mixed	242	0.946	1.054	0.306	0.095
Supportive	1,026	0.432	0.440	0.066	0.317
Not about the ad	9,172	0.344	0.216	0.135	0.231

Table 9: Agreement with labels assigned blind to model output

Measure	n	Agreement	Cohen's κ	95 % CI
RoBERTa polarity	80	72.5 %	0.588	[0.44, 0.72]
VADER	80	63.7 %	0.450	[0.29, 0.61]
gpt-4.1-mini stance	40	85.0 %	0.718	[0.47, 0.89]

and stance codes, and a shareable extract with a codebook defining every column. Results: Tables 1–8 as CSV, both specification curves, every statistic quoted here as JSON, both validation samples, and eleven figures at 200 dpi.

Environment caveats. The collection machine carries `torchvision` 0.2.0 against `torch` 2.6, which breaks transformers' model loading; `analysis/score.py` shims the two attributes it needs, and the shim is inert on a clean install. `sentence_transformers` is unusable for the same reason, so embeddings are computed directly through `transformers` with mean pooling and L2 normalisation, numerically equivalent to `SentenceTransformer.encode(normalize_embeddings=True)` for this model.

Ethics and data handling. Public Reddit content from a public archive; no authentication and no private data. Usernames are retained as posted in the working files; a salted-hash version is available and should replace them before any public deposit, preserving author counts and cross-posting detection without naming individuals. Removed and deleted comments carry no text in the archive, so content their authors withdrew is not reproduced — only the fact and timing of removal is retained.

Cost and compute. Collection and all local models ran on CPU (Intel i7-6700K, 16 GB) in about 3.5 hours. The LLM pass cost $8.90 for 16,221 units.

9 Limitations

1. **Tone is not stance.** The stance coding addresses this for the research question, but the tone measures themselves remain valence rather than position.
2. **Clustering.** r/MensLib's effective sample is closer to five or ten conversations than to 1,335 observations; for stance, 58 threads contain a positioned comment and five hold 81 % of them.
3. **Soft clusters.** Themes are interpretive groupings, silhouette 0.077.
4. **Model dependence.** Stance figures are conditional on `gpt-4.1-mini` and the published prompt. A different model or prompt would need revalidation; the pilot showed prompt wording moving κ by 0.20.
5. **Asymmetric populations.** 4,846 r/MensRights authors against 549 in r/MensLib, with time spans that diverge after 2019.
6. **Unobserved removed content.** 1,145 comments carry no text. The bounds in Table 3 quantify the consequence: it is binding for tone and not for stance.

Table 10: Pipeline

Step	Command	Runtime
1. Collect	`python scrape.py all MensLib / ... MensRights`	2.5 h
2. Build tables	`python summary.py`	1 min
3. Clean and QA	`python -m analysis.prepare_llm`	30 s
4. Tone and emotion	`python analysis/score.py all`	35 min
5. Stance, constructs	`python -m analysis.llm_code run`	25 min
6. Tone inference	`python analysis/compare.py`	6 min
7. Stance inference	`python -m analysis.stance_compare`	4 min
8. Themes	`python -m analysis.themes`	12 min
9. Figures	`python analysis/figures.py`	1 min
10. Tables	`python -m analysis.report_tables`	2 min